



Authorship Verification using Impostor Projections and Siamese Networks

69.1% ↑

Accuracy
baseline 59.2%

72.3% ↑

F1-Score
baseline 58.9%

82.0%

Genuine Detection
improved model

56.5%

Impostor Detection
improved model

1,616

Test Pairs
795 genuine · 821 impostor

+10.2%

Accuracy Gain
same model · no retraining

THE PROBLEM

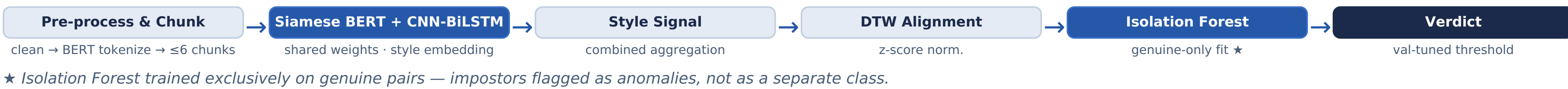
Authorship verification — determining whether two texts share the same author — is critical in **digital forensics**, **academic integrity**, **journalism**, and historical research.

- Traditional methods **fail against stylistic mimicry**, short texts, and topic shifts
- AI-generated content makes the problem **significantly harder**
- This project builds a framework that captures both **semantic meaning** and **stylistic signatures** simultaneously
- Model is hardened against imitation via the **Impostor Projection methodology**

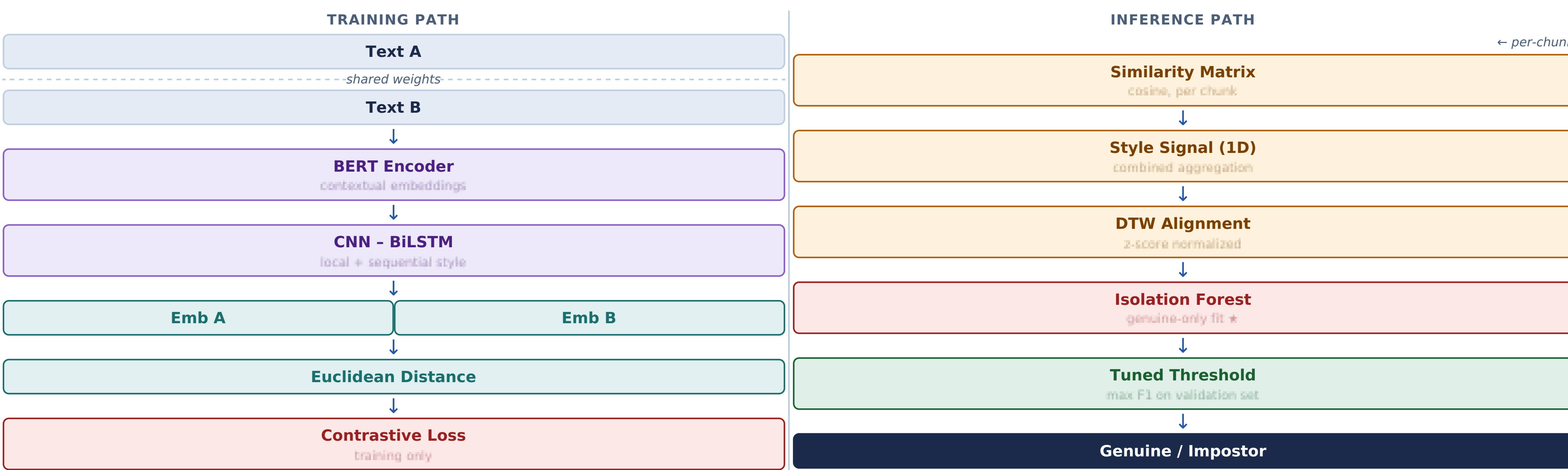
PROJECT REQUIREMENTS

- Accept **any two texts** — output genuine/impostor verdict + confidence score
- Handle **different lengths and topics** (no matched samples needed)
- Run on a single **free-tier 16GB GPU** (Tesla T4)
- Evaluated on **PAN20** — a public benchmark for fair comparison
- All settings in **one config module** for future reuse

METHODOLOGY — END-TO-END PIPELINE



ARCHITECTURE — SIAMESE BERT + CNN-BiLSTM



RESEARCH RESULTS — FINAL METRICS (PAN20 TEST SET)

Metric	Baseline mixed-fit · median split	Improved genuine-only · tuned threshold
Accuracy	59.2%	69.1% ↑
F1-Score	58.9%	72.3% ↑
Precision	58.4%	64.6% ↑
Recall	59.4%	82.0% ↑

Same trained model for both columns — improvement comes entirely from the decision strategy, with **no retraining**.

DETECTION RATES

Genuine detection 82.0%

Impostor detection 56.5%

DATASET — PAN20

PAN20 Benchmark 1,616 test pairs 795 genuine · 821 impostor Cross-topic · Variable length

CONFUSION MATRICES — BASELINE vs. IMPROVED

CONFUSION MATRICES — BASELINE vs. IMPROVED				
BASELINE				Actual Label
Predicted Label		Impostor (Imp)	Genuine (Gen)	
Actual Label	Impostor (Imp)	485	336	
	Genuine (Gen)	323	472	
Total		808	808	1,616
Accuracy: 59.2% F1-Score: 58.9%				
IMPROVED ✓				Actual Label
Predicted Label		Impostor (Imp)	Genuine (Gen)	
Actual Label	Impostor (Imp)	464	357	
	Genuine (Gen)	143	652	
Total		607	1,009	1,616
Accuracy: 69.1% F1-Score: 72.3%				



Genuine false negatives drop from 323 → 143 (~55.7%). The genuine-only detector learns a sharper authorship profile.

TRAINING DIAGNOSTICS

- Train loss** decreases steadily across 8 epochs — stable gradient flow confirmed
- Validation loss** minimum at **epoch 6** (val_loss = 0.1611) → early stopping triggered (patience = 2)
- Best checkpoint verified by **internal val_loss field**, not by filename (fixes known stale-checkpoint bug)
- LR schedule**: linear warmup → linear decay · Peak: 2×10^{-5} · Effective batch: $4 \times 8 = 32$

CHALLENGES & ENGINEERING SOLUTIONS

GPU Memory (CUDA OOM)

Two Siamese towers × many chunks exceeded 16GB. Fixed: **cap at 6 chunks/text** + gradient accumulation (effective batch = 32).

Biased Verdict Distribution

Detector predicted "genuine" for ~90% of pairs. Fixed: replaced K-Medoids split with **F1-tuned validation threshold**.

Checkpoint Tracking Bug

Two independent "best loss" trackers disagreed. Fixed: **unified into one rule**; epoch verified by internal val_loss value.

Dependency Conflicts

NumPy upgrade broke scikit-learn-extra; SymPy/PyTorch mismatch broke BERT. Fixed: **pinned exact compatible versions**.

Switching the Isolation Forest from a mixed-data fit to a **genuine-only fit**, with a validation-tuned threshold, raised Accuracy 59.2% → 69.1% and F1-Score 58.9% → 72.3% — on the **same trained model, without any retraining**. This confirms that the **decision strategy**, not just the architecture, drives real-world verification accuracy.